

music quality is measured. Digital music's sound quality is measured using these two values – bit depth and sampling rate.

Sampling rate is the number of times a sample of the audio signal is taken per second. The more the sampling rate, more the samples taken every second which mean more details will be recorded and you will have a richer and clearer sound with more nuances.

Bit depth represents the number of bits of information taken in each sample. This means that a higher bit depth will have a more dynamic range of sound. This basically denotes how much variance between soft or loud sounds can be. A higher bit depth leads to a higher bit rates – more number of bits per second – and bigger the file.

The most common format used today is MP3 which uses lossy data compression, making the file size very small and portable but also leading to a degradation in the quality and loss of information. CD quality audio noticeably sounds better and has a higher bit rate which in turn leads to a higher file size. High Resolution audio promises something better than this also.

What is high res audio and what is all the hype about?

If an audio file has both, a high sampling rate and a high bit rate (high being a relative term here), it will have more details and sounds more rich and realistic. But this is a very vague definition. How high is high enough to be called high resolution?

A number of major music labels including Universal Music Group and Warner Music group along with Song Electronics and Consumer Electronic association have agreed upon a standard for defining high-res audio. Any audio with a sampling rate higher than 44.1 kHz and a bit depth higher than 16 bit can be called a high res or high definition audio. Generally these files either have a 48 kHz/16 bit or 192 kHz/24 bit sampling rate and bit depth.

But what does all this mean? Does high res audio sound better than the music you rip off CDs? If it does, then how much is the difference between the two?

This is a difficult to question to answer. Technically speaking and in theory, yes, high res audio has more details than all other audio formats, but as many experts in the field argue, it is not possible for a everyone to tell the difference. The 44.1 kHz sampling rate and 16 bit depth, isn't some random number decided upon by companies. Near perfect sound is reproduced for human ears at these numbers.